

Introduction

1.4 Measures of Fluid Mass and Weight

Density

Density is defined as the mass per unit volume of a substance.

- BG system: $[\rho] = \frac{\text{slug}}{\text{ft}^3}$
- EE system: $[\rho] = \frac{\text{lbm}}{\text{ft}^3}$
- SI system: $[\rho] = \frac{\text{kg}}{\text{m}^3}$

Density of fluids are dependent on both temperature and pressure but for liquids, variations in temperature and pressure have a negligible effect on density. For gases, however, density is highly dependent on both temperature and pressure.

Specific Volume

Defined as volume per unit mass of a substance, and is the reciprocal of density:

$$v = \frac{1}{\rho} = \frac{V}{m}$$

More common in thermodynamics (see spring 2026 with KAMAU WRIGHT) than fluid mechanics.

Specific Weight

Defined as weight per unit volume of a substance:

$$\gamma = \rho g$$

Where g is the *local* acceleration of gravity:

- $g = 32.174 \frac{\text{ft}}{\text{s}^2}$ (BG)
- $g = 9.807 \frac{\text{m}}{\text{s}^2}$ (SI)

It has units of

- BG system: $[\gamma] = \frac{\text{lbf}}{\text{ft}^3}$
- SI system: $[\gamma] = \frac{\text{N}}{\text{m}^3}$

Specific Gravity

Defined as the ratio of fluid density to water at 4 °C or 39.2 °F:

$$\text{SG} = \frac{\rho}{\rho_{H_2O @ 4^\circ\text{C}}}$$

The density of water at room temperature, $\rho_{H_2O @ 4^\circ C}$ is:

- BG system: $\rho_{H_2O @ 4^\circ C} = 1.94 \frac{\text{slug}}{\text{ft}^3}$
- SI system: $\rho_{H_2O @ 4^\circ C} = 1000 \frac{\text{kg}}{\text{m}^3}$

Conversions:

- 1 slug = 32.174 lbm
- 1 lbf = 1 $\frac{\text{slug ft}}{\text{s}^2}$
- 1 slug = 14.5939 kg
- 1 m = 3.28084 ft

1.5 Ideal Gas Law (Equation of State)

$$p\vartheta = RT$$

Where:

- p : absolute pressure of the gas
- ϑ : specific volume of the gas
- R : Gas constant for the gas
- T : absolute temperature of the gas

Alternative form:

$$p = \rho RT$$

Constant Volume Process

For example, processes within a rigid container (V , R , and m are constant):

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Pressure

Pressure is defined as force per unit area and has units of

- BG system: $[\text{pressure}] = \frac{\text{lbf}}{\text{ft}^2}$ (psf)
- EE system: $[\text{pressure}] = \frac{\text{lbf}}{\text{in}^2}$ (psi)
- SI system: $[\text{pressure}] = \text{Pa} = \frac{\text{N}}{\text{m}^2}$ (Pa)

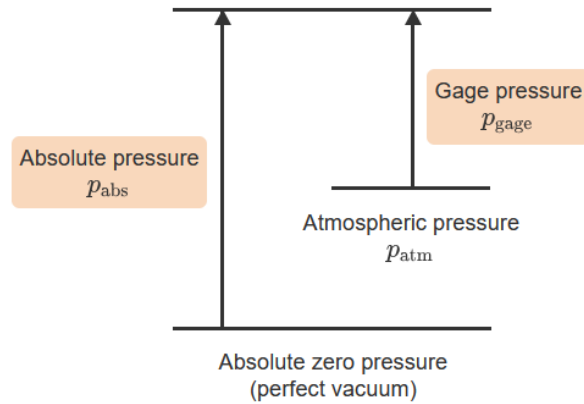
Absolute vs Gage Pressure

Measured relative to absolute zero pressure (perfect vacuum). While gage pressure is measured relative to atmospheric pressure. Thus, the relationship between absolute and gage pressure is:

$$(\text{Gage Pressure}) + (\text{Atmospheric Pressure}) = (\text{Absolute Pressure})$$

Where atmospheric pressure is:

- EE system: $p_{atm} = 14.7 \text{ psi}$
- SI system: $p_{atm} = 101 \text{ kPa}$



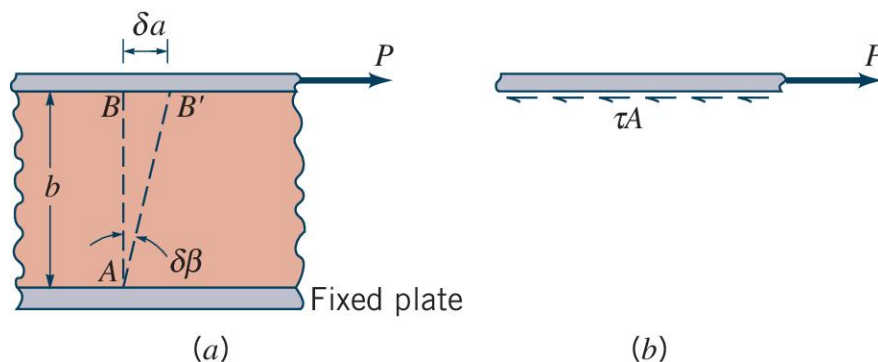
Conversions:

- $K = ^\circ C + 273.15$
- $R = ^\circ F + 460$

1.6 Viscosity

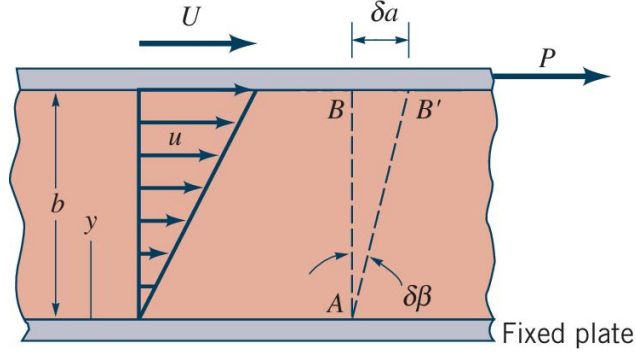
Viscosity describes the "fluidity" of a fluid, or how easily a fluid flows. It is a measure of the internal resistance of a fluid to flow. The higher the viscosity, the more resistant the fluid is to flow.

Consider the following set up with two parallel plates where the bottom plate is fixed, and the top plate is pulled by a load P . In between the plates is a solid (ie. steel) fixed to both plates.



Under the load P , the steel will deform, and the top plate will be displaced a small δa . AB will be rotated by a small angle $\delta\beta$ to AB' . We call $\delta\beta$ the **shearing strain**.

Now consider if a *fluid* is placed in between the plates instead of a solid. When a load P is applied, the top will now move continuously with a velocity U instead of stopping after a displacement δa . This is due to the property of fluids that they cannot resist a shearing stress, and thus they will **deform continuously** under a shearing stress.



The fluid in contact with the upper plate has the same velocity U and the fluid in contact with the lower plate has zero velocity. Thus, in between the plates, the fluid has a linear velocity profile:

$$u(y) = \frac{U}{b}y$$

The fluid effectively "sticks" to the solid boundaries, this is known as the **no-slip condition**.

The velocity gradient is then defined by:

$$\frac{du}{dy}$$

Rate of Shearing Strain

In a small time increment δt , the top plate moves a distance δa and the line AB rotates by a small angle $\delta\beta$. The shearing strain is then defined as:

$$\delta\beta \approx \tan \delta\beta = \frac{\delta a}{b} \quad (\text{by the small angle approx})$$

Since $\delta a = U\delta t$, we can express the shearing strain as:

$$\delta\beta = \frac{U\delta t}{b}$$

Notice that $\delta\beta$ is a function of the velocity U (which is dependent on P) and time.

The **rate of shearing strain** $\dot{\gamma}$ is then defined as the limit as $\delta t \rightarrow 0$:

$$\dot{\gamma} = \lim_{\delta t \rightarrow 0} \frac{\delta\beta}{\delta t}$$

For the linear velocity profile given in the example above, the rate of shearing strain is $\dot{\gamma} = \frac{U}{b} = \frac{du}{dy}$.

Viscosity (omg that's the section name)

Further experimentation would find that the shearing stress τ is proportional to the rate of shearing strain $\dot{\gamma}$. Thus, for common fluids (such as water, oil, gasoline, and air), we can state the relationship:

$$\tau = \mu \dot{\gamma}$$

Where μ is the **dynamic/absolute viscosity** of the fluid.

Units of Viscosity

The units of dynamic viscosity are FTL^{-2} :

- BG & EE system: $[\mu] = \frac{\text{lbf s}}{\text{ft}^2}$
- SI system: $[\mu] = \text{Pa s} = \frac{\text{N s}}{\text{m}^2}$

Types of Fluid

Different types of fluids can be classified based on the relationship between shearing stress and rate of shearing strain:

- **Newtonian Fluids** have a linear relationship between shearing stress and rate of shearing strain du/dy . (Ex: water, air, gasoline, and oil)
- **Shear-Thinning Fluids** have a non-linear relation with the rate of shearing strain, where the **apparent viscosity** decreases as shear rate increases. In other words, μ is not constant. (Ex: latex paint, not yummy...)
- **Shear-Thickening Fluids** have a non-linear relation with the rate of shearing strain, where the apparent viscosity increases as shear rate increases. (Ex: cornstarch in water, crunchy)
- **Bingham Plastics** behave like a solid at low shear stress, but flow like a fluid at high shear stress. It doesn't flow until yield stress is exceeded. (Ex: toothpaste, ketchup, and mayonnaise)

Effect of Temperature on Viscosity

The effect of temperature on viscosity can be approximated by the following empirical relationship called the **Sutherland Equation** for gases:

$$\mu = \frac{CT^{3/2}}{T + S}$$

Where C and S are constants for a given gas and T is the absolute temperature.

For **liquids**, the empirical relationship that is used is the **Andrade Equation**:

$$\mu = De^{B/T}$$

Where D and B are constants for a given liquid and T is the absolute temperature.

Kinematic Viscosity

The kinematic viscosity ν is defined as the ratio of dynamic viscosity to density. Often dynamic viscosity appears in fluid flow problems together with density, thus it is convenient to define a placeholder variable for this ratio:

$$\nu = \frac{\mu}{\rho}$$

It has units of L^2/T

Reynold's Number

This is a dimensionless number that describes the ratio of inertial forces to viscous forces in a fluid flow:

$$\text{Re} = \frac{\rho V D}{\mu}$$

Where

- ρ : density of the fluid
- V : mean velocity of the fluid
- D : characteristic length (such as diameter of a pipe)
- μ : dynamic viscosity of the fluid

1.7 Compressibility of Fluids

Bulk Modulus E_ν

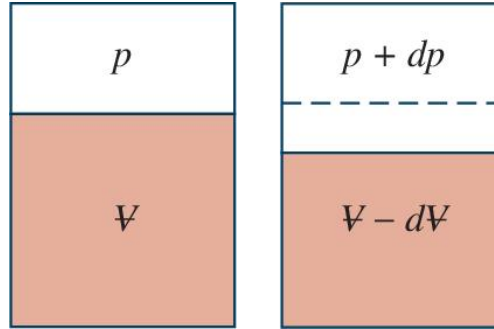
Describes how easily the volume of a fluid can be compressed. Keeping mass constant, this also describes how easily the density of a fluid can be changed under a change in pressure applied. It is defined as

$$E_\nu = -\frac{dp}{dV/V}$$

Or alternatively, in terms of density. Since $V = m/\rho$, a decrease in volume corresponds to an increase in density:

$$E_\nu = \frac{dp}{d\rho/\rho}$$

Where dp is the change in pressure to cause a dV change in volume, or a $d\rho$ change in density. The first equation is negative since an increase in pressure results in a decrease in volume.



Units

It has units of pressure FL^{-2}

For a large bulk modulus, the fluid is **incompressible**, and for a small bulk modulus, the fluid is **compressible**.

Gas Compression and Expansion Processes

Constant Temperature Process (Isothermal)

Derived from the ideal gas law:

$$\frac{p}{\rho} = \text{constant}$$

Constant Entropy Process (Isentropic)

For a frictionless process with no heat exchange with surroundings:

$$\frac{p}{\rho^k} = \text{constant}$$

Where k is the ratio of specific heats $k = c_p/c_v$. c_p is the specific heat at constant pressure, and c_v is the specific heat at constant volume. Additionally, the specific heats are related by the gas constant R :

$$R = c_p - c_v$$

Rewriting Bulk Modulus Relations

Given the above relations between density and pressure for isothermal and isentropic processes, we can rewrite the bulk modulus as:

$$E_\nu = p \quad (\text{Isothermal})$$

$$E_\nu = kp \quad (\text{Isentropic})$$

Note: Since the pressure and density relations were derived from the ideal gas law, these relations are only valid for ideal gases and pressures are always absolute pressures.

Speed of Sound c

As a result of the compressibility in fluids, localized pressure disturbances require non-zero time to propagate through the fluid and travels at a finite speed. This is known as the speed

of sound c and is related to the bulk modulus and density of the fluid:

$$c = \sqrt{\frac{dp}{d\rho}} = \sqrt{\frac{E_\nu}{\rho}}$$

Since the disturbance is a small change in pressure and density, there is negligible heat transfer so it can be approximated as an isentropic process where $p/\rho^k = \text{constant}$. Thus, the speed of sound can be expressed as:

$$c = \sqrt{\frac{kp}{\rho}} = \sqrt{kRT} \quad (\text{Ideal Gas Law})$$

Additionally, the **mach number** relates the speed of sound to an object moving through the fluid at velocity V :

$$\text{Ma} = \frac{V}{c}$$

1.9 Surface Tension

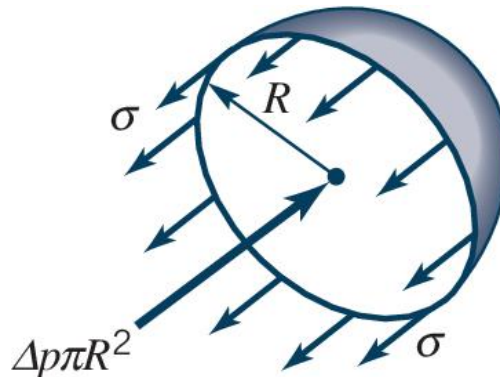
Surface tension exists at a liquid-gas interface.

Surface tension σ has units of force per unit length (FL^{-1}):

- BG system $[\sigma] = \frac{\text{lbf}}{\text{ft}}$
- SI system $[\sigma] = \frac{\text{N}}{\text{m}}$

Surface Tension and Pressure Within a Drop of Fluid

Consider the following free-body diagram of a cross-section of a spherical drop of fluid. The surface tension acts along the edge to shrink the drop's surface area while the pressure within the drop acts to expand the drop's surface area. At equilibrium, the two forces are equal:



Force due to surface tension: $2\pi R\sigma$

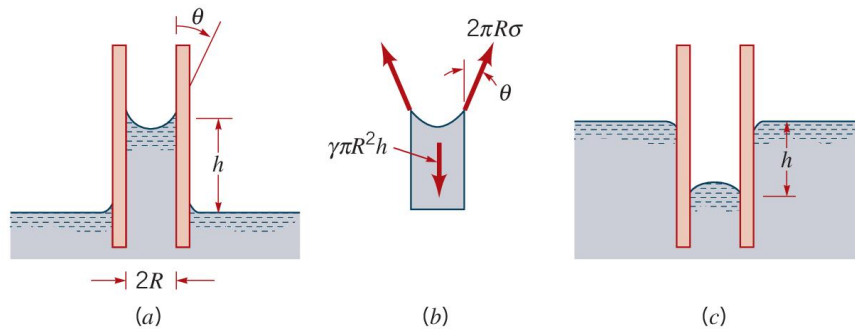
Force due to pressure within the drop: $\Delta p \pi R^2$ where Δp is the pressure difference across the drop's surface ($p_{\text{inside}} - p_{\text{outside}}$)

Setting the two forces equal:

$$2\pi R\sigma = \Delta p \pi R^2 \implies \Delta p = p_i - p_o = \frac{2\sigma}{R}$$

Capillary Action in small Tubes

In the figure below, (a) shows a rise in liquid level for a liquid that wets the tube (hydrophilic) and (c) shows a drop in liquid level for a nonwetting liquid (hydrophobic). The rise or drop in liquid level is due to the surface tension acting along the circumference of the tube. The free body diagram of the liquid for case (a) is shown in (b).



Since surface tension exists along the circumference, its magnitude is given by: $2\pi R\sigma$. With a vertical component of $2\pi R\sigma \cos \theta$ Where θ is the contact angle between the liquid and the tube.

The weight of the liquid column is given by: $\gamma \pi R^2 h$ where h is the height of the liquid column.

Setting the two forces equal at equilibrium to determine the height of the liquid column as a function of surface tension and tube geometry:

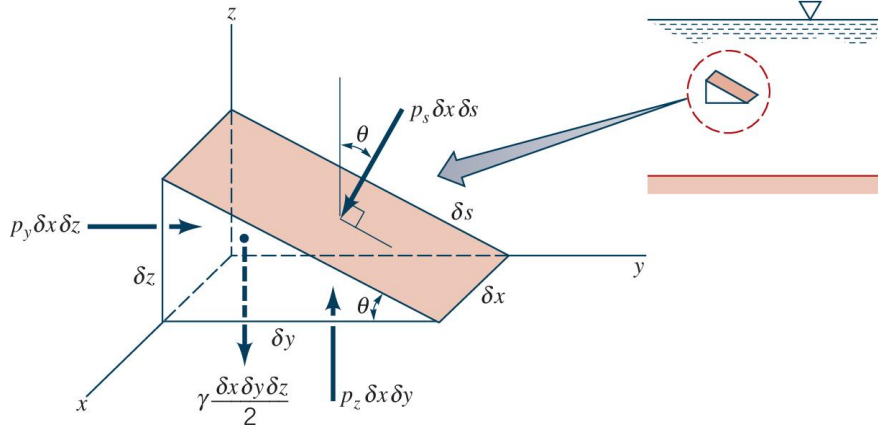
$$\begin{aligned} \sum F_y = 0 &= 2\pi R\sigma \cos \theta - \gamma \pi R^2 h \\ \implies h &= \frac{2\sigma \cos \theta}{\gamma R} \end{aligned}$$

Note that θ is dependent on both the liquid and tube material properties. $\theta \approx 0^\circ$ for water in clean glass.

2.1 Pressure at a Point

Pressure is defined as the normal force per unit area at a given point acting on a given plane. However, does the orientation of the plane matter? In other words, at the same point, does pressure depend on the orientation/direction of the plane?

Consider the following small triangular wedge of fluid mass. *Since there are no shearing stresses, the only external forces acting on the mass is pressure and gravity.* (Note that the forces in the x direction is not shown for simplicity)



Note that:

- The forces due to pressure are defined as the **average pressure in each direction** multiplied by the area of the surface in that direction. (Ex: $F_x = p_y A_y = p_y \delta x \delta z$)
- The weight of the fluid mass is given by $W = \gamma V = \gamma \left(\frac{\delta x \delta y \delta z}{2} \right)$. Where volume is the volume of the triangular wedge.
- The resultant force on the cut plane is given by $F_s = p_s A_s = p_s \delta s \delta x$. Where δs is the length of the cut plane.

By Newton's second law in the y and z directions, we can write the following equations:

$$\sum F_y = p_y \delta z \delta x - p_s \delta x \delta s \sin \theta = \rho \left(\frac{\delta x \delta y \delta z}{2} \right) a_y$$

$$\sum F_z = p_z \delta x \delta y - p_s \delta x \delta s \cos \theta - \gamma \left(\frac{\delta x \delta y \delta z}{2} \right) = \rho \left(\frac{\delta x \delta y \delta z}{2} \right) a_z$$

It follows from geometry that

$$\delta y = \delta s \cos \theta, \quad \delta z = \delta s \sin \theta$$

So the equations of motion is rewritten as:

$$p_y - p_s = \rho a_y \frac{\delta y}{2}$$

$$p_z - p_s = (\rho a_z + \gamma) \frac{\delta z}{2}$$

To determine the pressure distribution at a point, we take the limit as $\delta y, \delta z \rightarrow 0$, keeping θ constant. Thus, the right-hand side of both equations goes to zero, and we get:

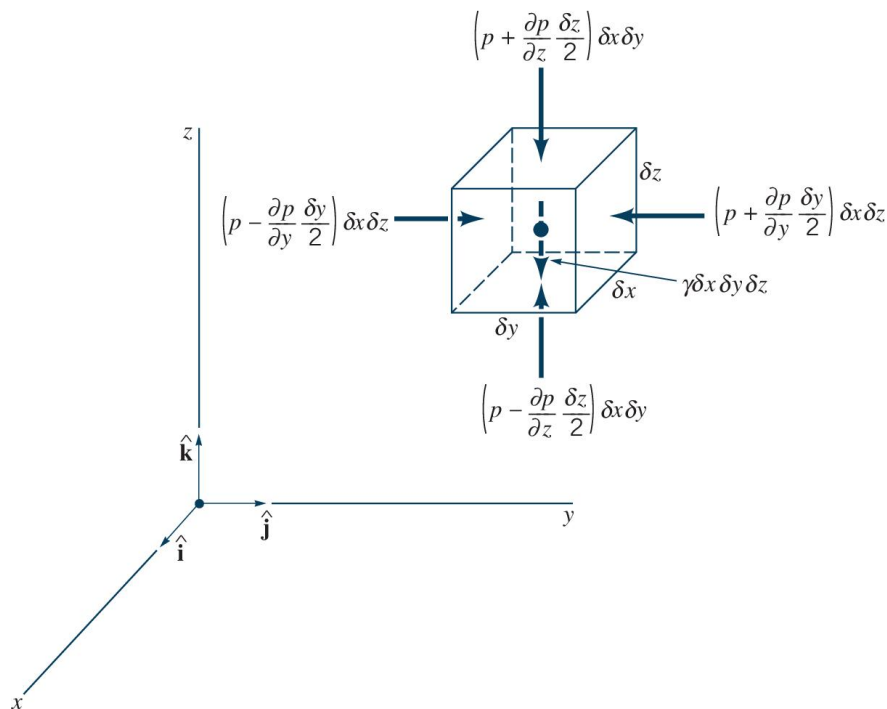
$$p_y = p_s, \quad p_z = p_s \implies p_x = p_y = p_z \quad (\text{if we had also considered } p_x)$$

Therefore, since θ is arbitrary, the pressure at a point in a fluid at rest, or in motion, is independent of direction as long as there are no shearing stresses present. This is known as **Pascal's Law**.

Note: The version of Pascal's law that states that *a change in pressure at any point in a continuous fluid is transmitted to all points in the fluid* is mainly a result of the hydrostatic pressure distribution when we neglect elevation differences across a fluid. (Same Pascal, different law?)

2.2 Basic Equation for Pressure Field

Now that we know that pressure is independent of direction at a point, we should now ask how pressure varies across points in a fluid (ie. find the pressure distribution). Consider a small rectangular fluid element with dimensions $\delta x, \delta y, \delta z$ in a fluid without shearing stresses.



For simplicity, the forces in the x direction are not shown.

There are two main forces acting on the element: surface forces due to pressure, and body force due to weight.

We define the pressure at the center of the element (x, y, z) as p . The average pressure on each face is approximated using a first-order Taylor expansion of p about the center of the element, evaluated a distance $\delta/2$ away in the corresponding coordinate direction.

The sum of surface forces in the y direction is

$$\delta F_y = \left(p - \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \delta z - \left(p + \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \delta z$$

or

$$\delta F_y = -\frac{\partial p}{\partial y} \delta x \delta y \delta z$$

Similarly, for the x and z directions, the sum of surface forces are

$$\delta F_x = -\frac{\partial p}{\partial x} \delta x \delta y \delta z, \quad \delta F_z = -\frac{\partial p}{\partial z} \delta x \delta y \delta z$$

The resultant surface force acting on the element due to pressure can be expressed in vector form as

$$\delta \vec{F}_s = \delta F_x \hat{i} + \delta F_y \hat{j} + \delta F_z \hat{k}$$

or

$$\delta \vec{F}_s = -\left(\frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{k} \right) \delta x \delta y \delta z$$

where $\hat{i}$, $\hat{j}$, and $\hat{k}$ are the unit vectors along the coordinate axes.

This can be rewritten in terms of the **gradient operator** ∇ as:

$$\frac{\delta \vec{F}_s}{\delta x \delta y \delta z} = -\nabla p, \quad \nabla p = \frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{k}$$

Combining Surface Forces with Body Forces

To describe all the forces acting on the element, we need to include the body force due to weight. This is given by:

$$-\delta \mathcal{W} \hat{k} = -\gamma \delta x \delta y \delta z \hat{k}$$

The total force acting on the element is then given by:

$$\sum \delta \vec{F} = \delta \vec{F}_s - \delta \mathcal{W} \hat{k} = \delta m \vec{a}, \quad \text{where } \delta m = \rho \delta x \delta y \delta z$$

$$\implies \boxed{-\nabla p - \gamma \hat{k} = \rho \vec{a}}$$

This is the general equation of motion for a fluid with no shearing stresses. Note that for a fluid at rest, $\vec{a} = 0$, and the equation reduces to:

$$\nabla p = -\gamma \hat{k} \implies \frac{\partial p}{\partial z} = -\gamma$$

Which once integrated, gives the hydrostatic pressure distribution:

$$p = -\gamma z + p_0$$

Where z is defined as upwards positive.

2.3 Pressure Variation at Rest

For a fluid at rest, $\vec{a} = 0$, and the equation of motion reduces to:

$$-\nabla p - \gamma \hat{k} = 0$$

$$\implies \frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = -\gamma$$

Thus, pressure only changes with depth in the z direction.

Incompressible Fluid

By definition, $\gamma = \rho g$, since changes in g is negligible for most cases, the main factor that affects γ is the density ρ . For an incompressible fluid, ρ is constant.

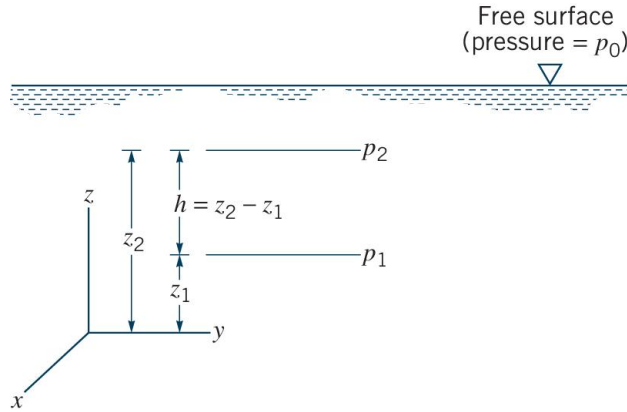
For a constant ρ , and thus γ , we can integrate the equation for pressure variation with depth:

$$\int_{p_1}^{p_2} dp = \int_{z_1}^{z_2} -\gamma dz$$

$$\implies p_2 - p_1 = -\gamma(z_2 - z_1)$$

If we let $h = z_2 - z_1$, then we can express the pressure at a depth h below a reference point as:

$$p_1 = p_2 + \gamma h$$



This is called a **hydrostatic distribution**.

Pressure Head

We can rewrite the hydrostatic distribution in terms of a height of a fluid column of specific weight γ and height h that would produce the same pressure difference:

$$h = \frac{p_1 - p_2}{\gamma}$$

Note: Often, we use the free surface of the fluid with reference pressure p_0 , and thus set $p_2 = p_0$ to find pressures at depth h below the free surface.

Compressible Fluids

Compressible fluids such as gases tend to have varying density when variations in heights/elevation is very large. Thus, combining the ideal gas law ($p = \rho RT$) with the equation of motion for a fluid at rest ($\partial p / \partial z = -\gamma$), we can derive the pressure distribution for a compressible fluid.

$$\frac{dp}{dz} = -\gamma = -\rho g = -\frac{pg}{RT}$$

Integrating both sides, assuming g and R are constant,

$$\int_{p_1}^{p_2} \frac{dp}{p} = -\frac{g}{R} \int_{z_1}^{z_2} \frac{1}{T} dz$$

Before solving the integral, we must consider how temperature varies in z !

Isothermal Process

For an isothermal process, $T = T_0$ is constant, and the integral can be solved as:

$$p_2 = p_1 \exp \left[-\frac{g(z_2 - z_1)}{RT_0} \right]$$

2.4 Standard Atmosphere

The variation of temperature within the troposphere (extends to an altitude of about 11 km or $\approx 36,000$ ft) is of the form

$$T = T_a - \beta z$$

Where

- T_a is the temperature at sea level ($z = 0$)
- β is the temperature lapse rate (rate of change of ΔT with respect to Δz). For standard atmosphere in the troposphere, $\beta = 0.0065 \frac{\text{K}}{\text{m}}$ or $0.00357 \frac{\text{R}}{\text{ft}}$

Integrating the equation for pressure variation with depth for a compressible fluid from 2.3, we have:

$$\int_{p_a}^p \frac{dp}{p} = -\frac{g}{R} \int_0^z \frac{1}{T} dz$$
$$p = p_a \left(1 - \frac{\beta z}{T_a} \right)^{g/(R\beta)}$$

For varying temperature in the troposphere.

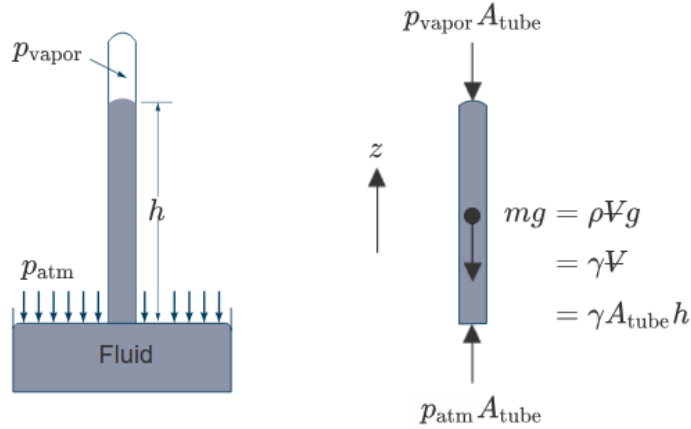
2.5 Measurement of Pressure

Mentioned before in 1.5, absolute pressure is relative to vacuum (absolute zero), gage is relative to local atmospheric pressure, and vacuum/suction is a negative gage pressure.

$$p_{abs} = p_{gage} + p_{atm}$$

Barometer

A way to measure atmospheric pressure using a column of liquid in terms of head (ex: mmHg) is by using a barometer. Consider the following barometer with a column of mercury.



For the mercury column to be at equilibrium, the pressure at the bottom of the column must be equal to the atmospheric pressure:

$$p_{atm} = \gamma h + p_{vapor}$$

Note: The vapor pressures can be found in tables 1.5 and 1.6 for different liquids at different temperatures.

This can be derived by a force balance on the mercury column:

$$\sum F_z = p_{atm} A_{tube} - \gamma A_{tube} h - p_{vapor} A_{tube} = 0$$

$$\implies p_{atm} = \gamma h + p_{vapor}$$

Assuming the vapor pressure is negligible: $p_{atm} \approx \gamma h$.

2.6 Man-O-Meters

Oh boy my favorite.

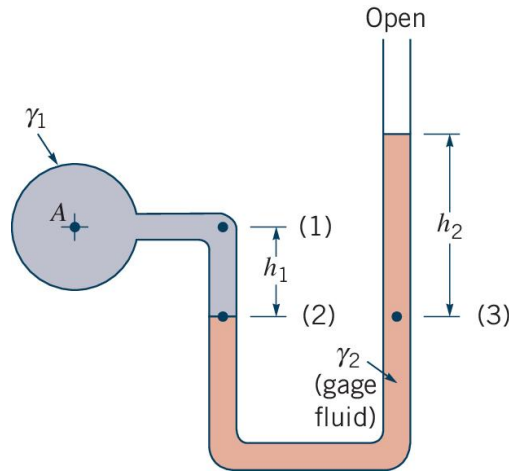
Man-O-Meters or manometers uses vertical or inclined liquid columns to measure pressure based on the same principle equation as the barometer $p = \gamma h + p_0$.

To solve manometer problems, we can use the following "rules":

1. Write pressure at either end of manometer
2. Move through manometer and add γh with increasing depth and subtract γh with decreasing depth
3. When stopping at any point, the sum is equal to the local pressure at that point

Make horizontal jumps without adding since the pressure is constant along a horizontal line.

For example, the following is a man-ohh-meter:

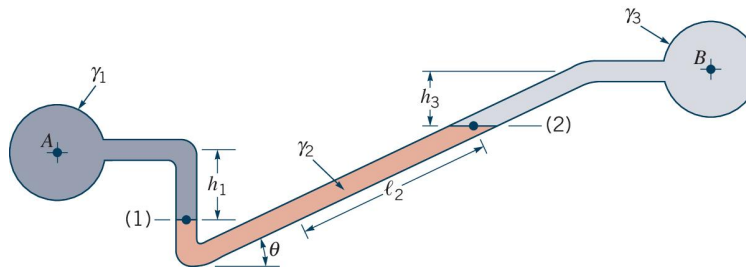


If we start from the open end:

$$p_{atm} + \gamma_2 h_2 - \gamma_1 h_1 = p_A$$

Inclined-tube Manometer

If we want to be fancy and physically add more resolution to our manometer, we can incline the tube. This allows for a longer column of liquid to measure the same pressure difference. See the following example:



The pressure difference $p_A - p_B$ can be determined by moving through the manometer starting from p_A :

$$\begin{aligned} p_A + \gamma_1 h_1 - \gamma_2 \ell_2 \sin \theta - \gamma_3 h_3 &= p_B \\ \Rightarrow p_A - p_B &= \gamma_2 \ell_2 \sin \theta + \gamma_3 h_3 - \gamma_1 h_1 \end{aligned}$$

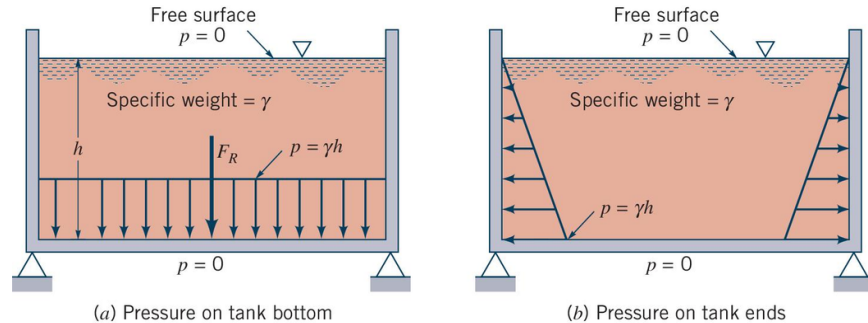
The same manometer rules applies but be careful to use the vertical height instead of the length of the inclined tube. The vertical height is given by $\ell_2 \sin \theta$.

Hydrostatics

2.8 Hydrostatic force on a plane surface

For fluid at rest, fluid pressure acts perpendicular to any surface because there is no shear stress present in a static fluid.

For example, the pressure and force distribution on an open tank is shown below:



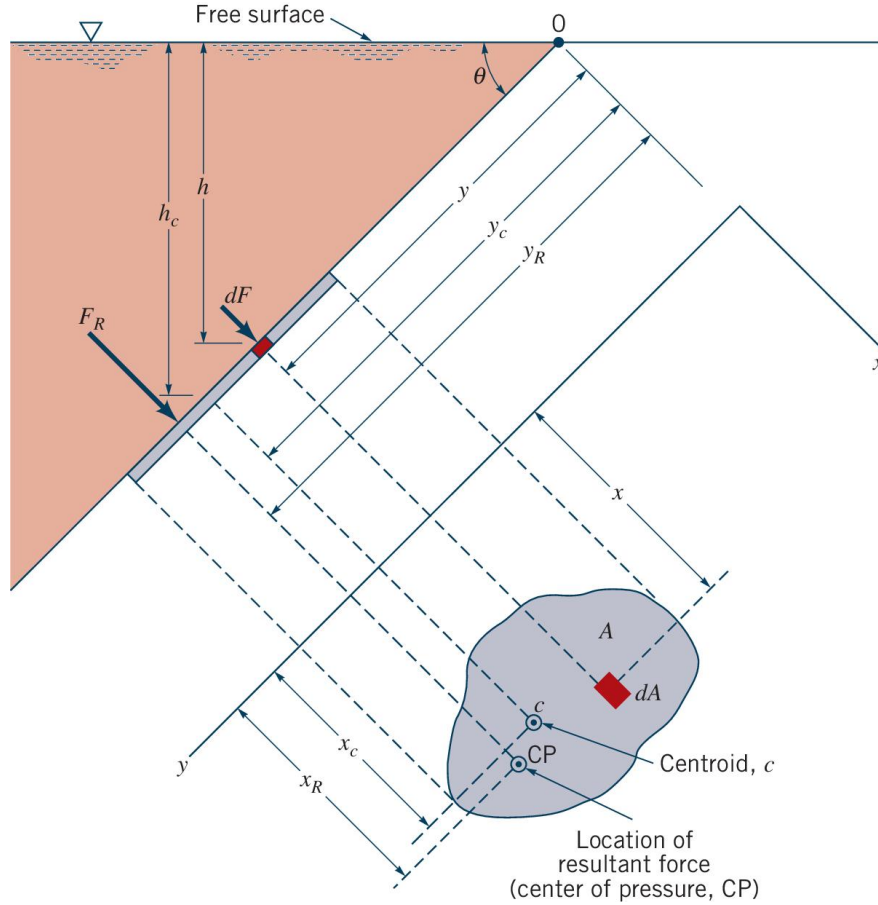
Note that if atmospheric pressure acts on both sides and bottom of the tank, it will cancel out and can be ignored (which is why we often use gage pressure). The resultant force on the bottom of tank is given by:

$$F_R = pA, \quad \text{where } p = \gamma h$$

Since the pressure is constant and uniformly distributed over the bottom, the resultant force acts through the **centroid of the area**. However, for the sides of the tank the pressure varies with depth, so it's harder to determine the resultant force and its line of action.

Resultant Force on a Plane Surface

Consider the following submerged inclined plane surface below. The free surface is open to the atmosphere. We define a xy coordinate system so that the surface lies on the xy plane, with the origin where the plane in which the surface lies intersect the free surface.



The force due to the fluid pressure in contact with the surface is given by integrating the pressure over the area of the surface:

$$F_R = \int_A p dA = \int_A \gamma h dA$$

Where h is the depth of the surface element dA below the free surface (h is in the vertical direction, NOT in the y direction). Note that the depth h can be expressed in terms of the y coordinate by the angle of inclination θ :

$$h = y \sin \theta$$

Also note that the integral for F_R can be expressed in terms of the first moment of area of the surface about the x -axis:

$$Q = \int_A y dA = A\bar{y} \quad (\text{First Moment of Area})$$

Where y is the distance from the x -axis to the area element dA , and $\bar{y}$ is the distance from the x -axis to the centroid of the area. Thus, using some new notation, we can express the resultant force as:

$$F_R = \int_A \gamma h dA = \int_A \gamma y \sin \theta dA$$

For a constant γ and θ , and $h = y \sin \theta$:

$$F_R = \gamma \sin \theta \int_A y dA = \gamma \sin \theta A y_c$$

Where $\bar{y} = y_c$ is the distance from the x -axis to the centroid of the area along the y -axis. Finally:

$$\boxed{F_R = \gamma h_c A}$$

y Coordinate of the Resultant Force

Recall that the resultant force acts through the centroid of the area when the pressure is constant (like the bottom of the tank). However, when the plane surface is inclined, the pressure increases with depth, and the resultant force acts slightly below the centroid of the area.

We can determine the y coordinate of the resultant force (y_R) by summing moments about the x -axis. The moment of the resultant force at y_R is equal to the moment of the distributed pressure about the x -axis:

$$F_R y_R = \int_A y dF, \quad dF = \gamma h dA = \gamma y \sin \theta dA$$

$$\implies F_R y_R = \int_A \gamma \sin \theta y^2 dA$$

From before, $F_R = \gamma h_c A = \gamma \sin \theta A y_c$. So for a constant γ and θ :

$$\gamma \sin \theta A y_c y_R = \gamma \sin \theta \int_A y^2 dA$$

$$\implies y_R = \frac{\int_A y^2 dA}{y_c A}$$

Note that the integral in the numerator is the second moment of area about the x -axis, which is also known as the *area moment of inertia*:

$$I_x = \int_A y^2 dA$$

Thus, we can express the y coordinate of the resultant force as:

$$y_R = \frac{I_x}{y_c A}$$

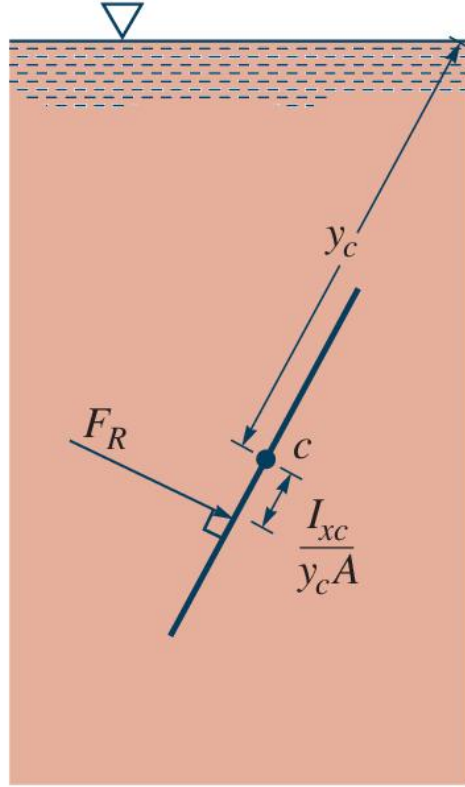
Using the parallel axis theorem, we can express I_x in terms of an axis parallel to the x -axis that passes through the centroid of the area:

$$I_x = I_{xc} + Ay_c^2$$

Thus,

$$y_R = \frac{I_{xc}}{y_c A} + y_c$$

Following the intuition from the beginning, since $I_{xc}/(y_c A)$ is always positive, the resultant force will always act below the centroid of the area.



Note: y_c is measured from the line of intersection of the xy plane with the surface, defining the angle θ . However, when the xy plane is horizontal and parallel to the free surface, $\theta \rightarrow 0^\circ$, and $y_c = h_c / \sin \theta \rightarrow \infty$. In this case, the term $I_{xc}/(y_c A) \rightarrow 0$, and the resultant force acts through the centroid of the area, as expected.

x Coordinate of the Resultant Force

Similarly, we can determine the x coordinate of the resultant force (x_R) by summing moments about the y -axis. The moment of the resultant force at x_R is equal to the moment of the distributed pressure about the y -axis:

$$F_R x_R = \int_A x dF, \quad dF = \gamma h dA = \gamma y \sin \theta dA$$

$$\implies F_R x_R = \int_A \gamma \sin \theta xy dA$$

Since $F_R = \gamma h_c A = \gamma \sin \theta A y_c$, for a constant γ and θ , we can divide through by F_R to get:

$$x_R = \frac{\gamma \sin \theta \int_A xy dA}{\gamma \sin \theta A y_c} = \frac{I_{xy}}{y_c A}$$

Like before, I_{xy} is the product of inertia about the x and y axes. Using the parallel axis theorem, we get the general expression for the x coordinate of the resultant force:

$$x_R = \frac{I_{xyc}}{y_c A} + x_c$$

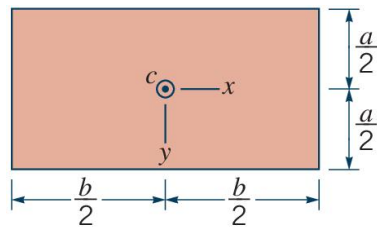
Where I_{xyc} is the product of inertia about the x and y axes that pass through the centroid of the area, and x_c is the x coordinate of the centroid of the area.

Note that if the plane surface is symmetrical about an axis parallel to either the x or y axis and passing through the centroid of the area, then $I_{xyc} = 0$, and $x_R = x_c$.

Together, x_R and y_R define the *center of pressure*, which is the point where the resultant force acts on the plane surface.

Note: For both x_R and y_R , the center of pressure will move closer to the centroid of the area as the plane surface becomes more horizontal ($\theta \downarrow$), and thus increases y_c in the denominator of both expressions. In the limit as $\theta \rightarrow 0^\circ$, $y_c \rightarrow \infty$, and the center of pressure coincides with the centroid of the area, as expected.

Common Area Moments of Inertia



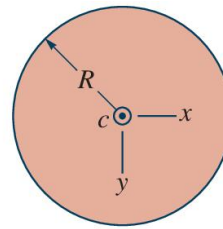
(a) Rectangle

$$A = ba$$

$$I_{xc} = \frac{1}{12} ba^3$$

$$I_{yc} = \frac{1}{12} ab^3$$

$$I_{xyc} = 0$$

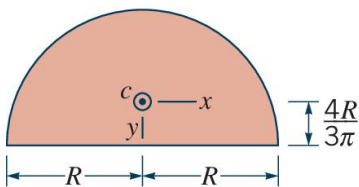


(b) Circle

$$A = \pi R^2$$

$$I_{xc} = I_{yc} = \frac{\pi R^4}{4}$$

$$I_{xyc} = 0$$



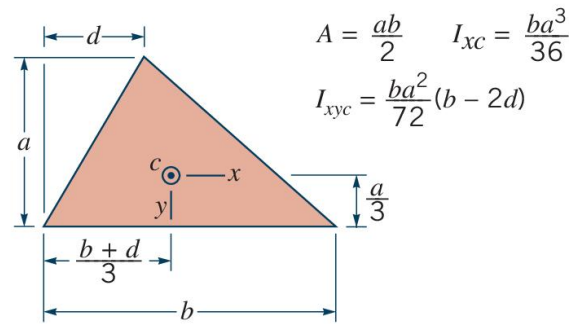
(c) Semicircle

$$A = \frac{\pi R^2}{2}$$

$$I_{xc} = 0.1098R^4$$

$$I_{yc} = 0.3927R^4$$

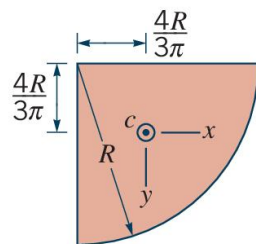
$$I_{xyc} = 0$$



(d) Triangle

$$A = \frac{ab}{2} \quad I_{xc} = \frac{ba^3}{36}$$

$$I_{xyc} = \frac{ba^2}{72}(b - 2d)$$



(e) Quarter circle

$$A = \frac{\pi R^2}{4}$$

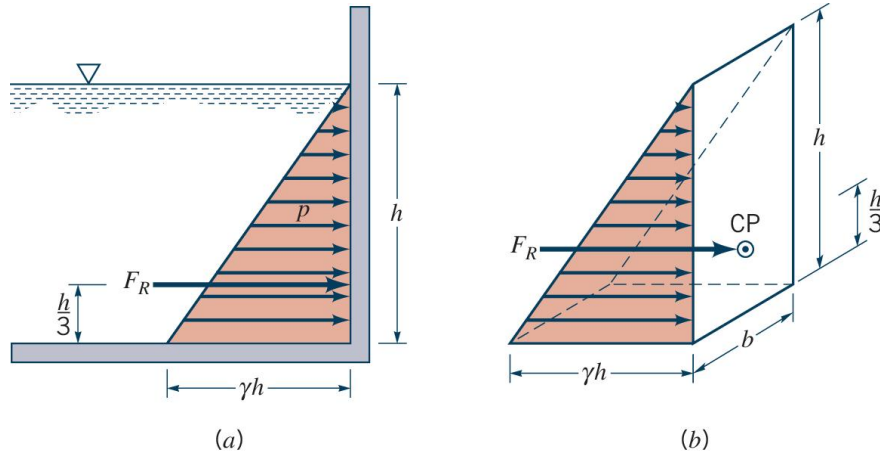
$$I_{xc} = I_{yc} = 0.05488R^4$$

$$I_{xyc} = -0.01647R^4$$

2.9 Pressure Prism

This section doesn't introduce any new concepts, but rather provides a graphical approach to solving for the resultant force and center of pressure on a plane surface. This approach is often convenient for rectangular surfaces, where the centroid and volume of the pressure prism can be easily determined.

Consider the following pressure prism on a vertical plane surface with constant width b :



Since the pressure varies linearly with depth, it is zero at the free surface ($p_{min} = 0$) and maximum at the bottom of the surface ($p_{max} = \gamma h$). Thus, the average pressure on the surface is $p_{avg} = \frac{p_{min} + p_{max}}{2} = \frac{\gamma h}{2}$. The resultant force on the surface is given by:

$$F_R = p_{avg}A = \gamma \left(\frac{h}{2} \right) A$$

This is the same result as the equation $F_R = \gamma h_c A$ in section 2.8.

By forming a *pressure prism* (shown in figure (b)), the resultant force can be determined graphically as the **volume of the pressure prism**:

$$F_R = \text{volume} = \frac{1}{2}(\gamma h)(bh)$$

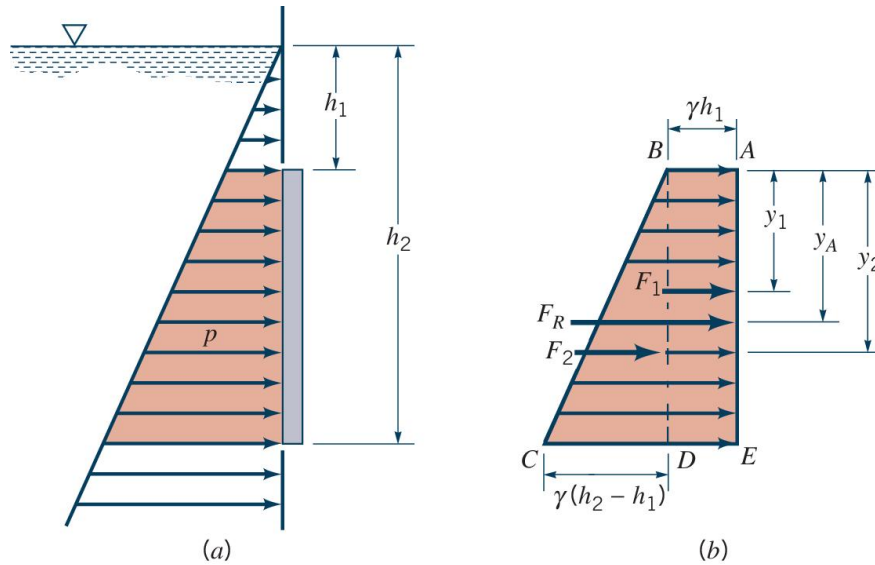
This is the same result as before as well.

The center of pressure can also be determined graphically as the **centroid of the pressure prism**. The centroid of a triangle is located at a distance of $\frac{1}{3}$ from the base, so the center of pressure is located at a distance of $\frac{h}{3}$ from the bottom of the surface. This is the same result as the equation $y_R = I_{xc}/(y_c A) + y_c$ in section 2.8.

Note: The pressure prism is always drawn starting from the free surface.

Rectangular Surfaces that do not intersect the free surface

Consider the following plane surface that does not intersect the free surface:



The pressure prism approach can still be used but in this case, it is trapezoidal instead of triangular. The resultant force can be found by decomposing the trapezoid into a rectangle and triangle, $ABDE$ and BCD , respectively.

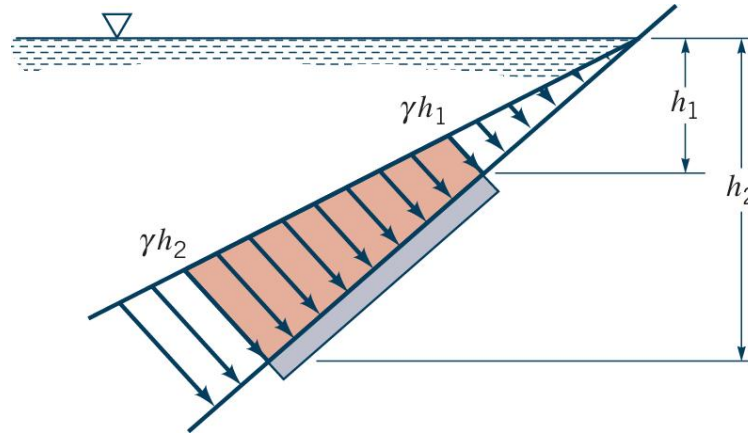
$$F_R = F_1 + F_2$$

The locations of the forces F_1 and F_2 can be found by taking the centroid of each shape, getting y_1 and y_2 , respectively. The location of the resultant force can then be found by summing moments about a convenient axis (in this case, the axis passing through A):

$$F_R y_R = F_1 y_1 + F_2 y_2$$

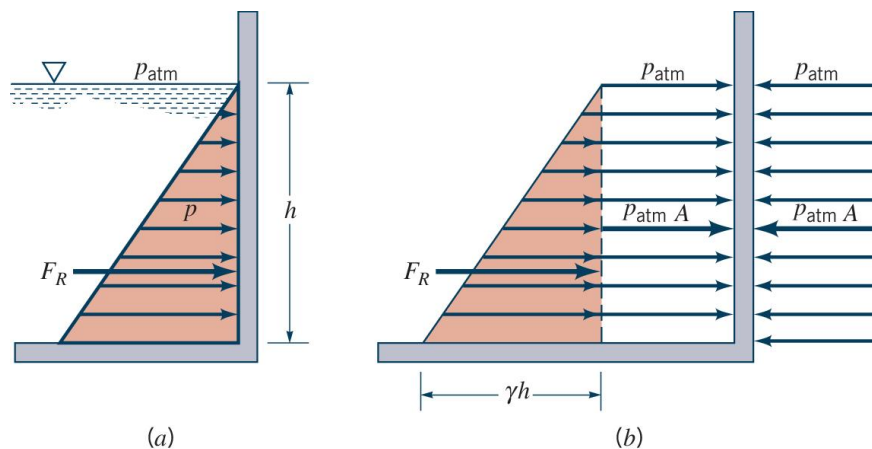
Inclined Rectangular Surfaces

A similar approach using trapezoidal pressure prisms can be used for inclined rectangular surfaces. However, note that while it is convenient to measure distances along the incline, the pressure varies with depth in the vertical direction.



Effect of Pressure at Free Surface

When we consider the effect of atmospheric pressure at the free surface, it simply offsets the pressure prism by a constant amount, resulting in a trapezoidal pressure prism.



In cases of open tanks, the atmospheric pressure acts on both sides of the vertical walls and so it cancels out. However, in cases where the pressure at the free surface is not equal to the pressure on the exterior surface, its effect is non-negligible.

Anyway, the analysis follows the same approach as the trapezoidal pressure prism.

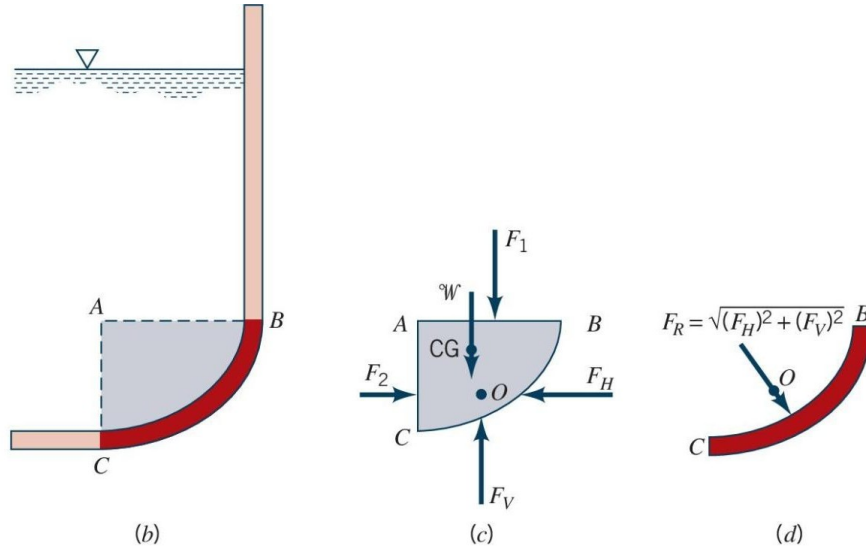
In summary, the volume of the pressure prism is basically:

$$(\text{Average pressure}) \cdot (\text{Area in contact with fluid})$$

2.10 Hydrostatic force on a curved surface

So far, we have only considered plane surfaces. What about curved surfaces? The resultant force on a curved surface can be determined by decomposing the force into horizontal and vertical components.

For example, consider the following curved surface BC :



Start by isolating a volume of fluid that is bounded by the surface of interest (b). Next, we determine the forces in the free-body diagram of the isolated fluid (c).

- F_1 : Force by fluid pressure on the *planar surface* AB . This can be determined using the methods from sections 2.8 and 2.9.
- F_2 : Force by fluid pressure on the *planar surface* AC . This can also be determined using the methods from sections 2.8 and 2.9.
- W : **Weight of the enclosed fluid volume**. This can be determined by multiplying the volume of the fluid by its specific weight and acts through the center of gravity of the volume.
- F_H and F_V : **The horizontal and vertical reaction forces from the tank on the fluid.**

Since the fluid is at static equilibrium, the sum of forces in both the horizontal and vertical directions must be zero.

$$\sum F_{\text{horiz}} = 0 = F_2 - F_H \implies F_H = F_2$$

$$\sum F_{\text{vert}} = 0 = F_1 + W - F_V \implies F_V = F_1 + W$$

Additionally, **for a body under three non-parallel forces to be in equilibrium, the lines of action of the three forces must intersect at a common point** (otherwise there would be a

non-zero moment). Therefore, we can say that the resultant force F_R passes through the common intersection point of the lines of action of F_H and F_V , O .

$$F_R = \sqrt{F_H^2 + F_V^2}$$

O can be determined by summing moments to find the lines of action of F_H and F_V , and then finding their intersection.

Note: The vertical component of the resultant force can actually be solved two ways:

1. Instead of isolating a small fluid region around the surface, we can isolate the entire fluid column above it. Then, the vertical hydrostatic force on the surface is just the weight of the fluid column above it:

$$F_V = \mathcal{W} = \gamma \mathcal{V}$$

2. Or alternatively, the procedure described before can be used, where we isolate a local fluid region around the surface. In that case, the vertical hydrostatic force on the surface is the sum of the weight of the fluid region and the vertical component of the hydrostatic force:

$$F_V = F_1 + \mathcal{W}$$

2.11 Buoyancy, flotation, and stability

Archimedes' Principle

The buoyant force on a body submerged in a fluid is equal to the weight of the fluid displaced by the body.

$$F_{\text{Buoyant}} = \gamma \mathcal{V}$$

Line of Action for the Buoyant Force

The point through which the buoyant force acts is called the *center of buoyancy*. The buoyant force passes through the centroid of the displaced volume.

For partially submerged bodies, if the specific weight of the fluid above the liquid surface is very small compared to the displaced fluid, the center of buoyancy can be approximated as the centroid of the submerged volume. This is a common approximation when the fluid above the surface is air.

If a body is submerged in a fluid of varying specific weight, such as a layered fluid, the magnitude of the buoyant force remains equal to the weight of the displaced fluid. However, the center of buoyancy is located at the center of gravity of the displaced fluid.

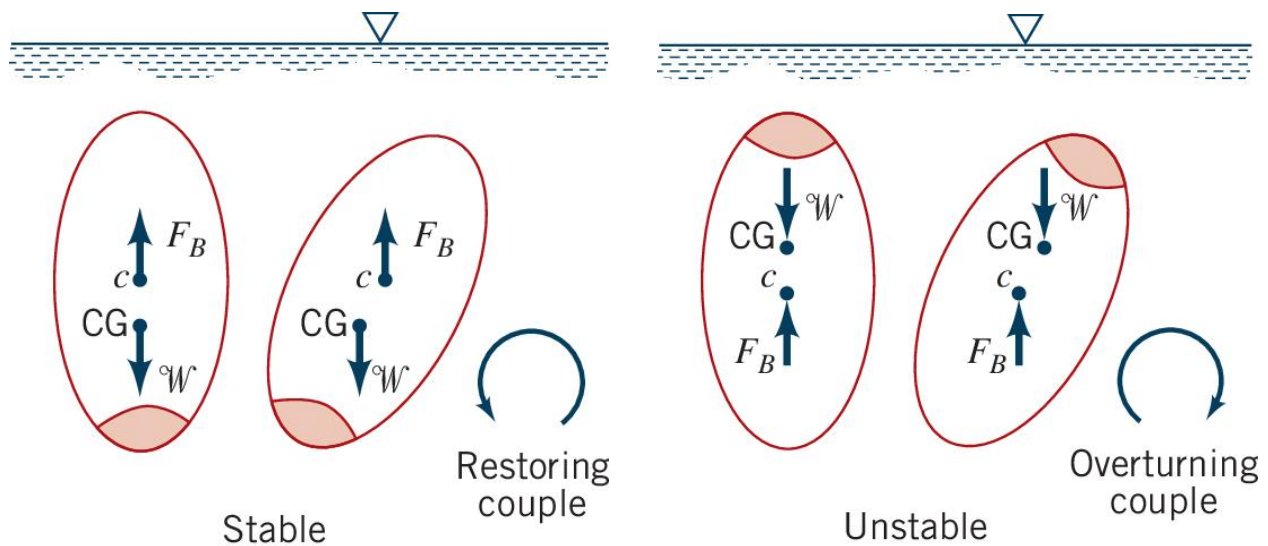
Note: The buoyant force is a special application of the hydrostatic pressure forces acting on the surfaces of a body. In other words, both buoyancy and hydrostatic pressure forces arise from the same physical phenomenon. Therefore, when performing calculations, use

either the resultant buoyant force or the individual hydrostatic pressure forces, but do not include both.

Stability

A body is said to be in stable equilibrium if it returns to its original position after being slightly disturbed, and in unstable equilibrium if it moves to a new equilibrium position instead.

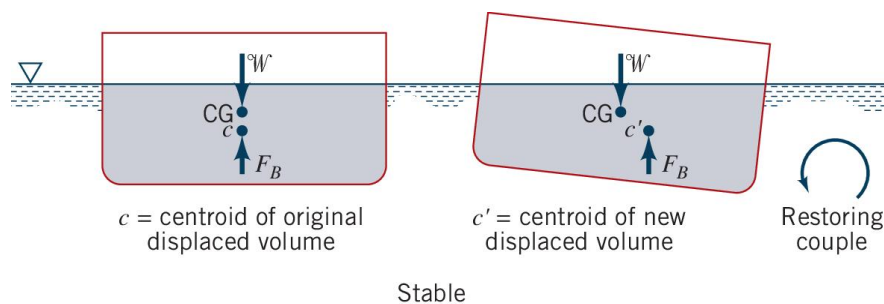
Stability of a **submerged body** can be determined by comparing the center of gravity of the body and the center of buoyancy:



The body is in stable equilibrium if the center of gravity is below the center of buoyancy, and in unstable equilibrium if the center of gravity is above the center of buoyancy.

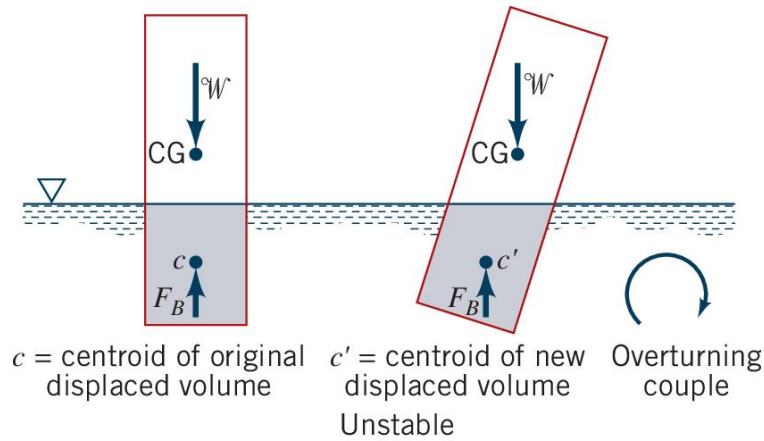
Floating Body Stability

For a floating body, the center of buoyancy (which is approximately the centroid of the submerged volume) *moves as the body is tilted*. Thus, it is possible for a wide floating body to be in stable equilibrium even if its center of gravity is above the center of buoyancy as seen below:



However, a narrow floating body may be in unstable equilibrium if its center of gravity

is above the center of buoyancy since the center of buoyancy may not move enough to counteract the tilt.



2.12 Pressure variation in a fluid with rigid-body motion

Recall that the general equation of motion for a fluid with no shearing stresses is given by:

$$-\nabla p - \gamma \hat{k} = \rho \vec{a}$$

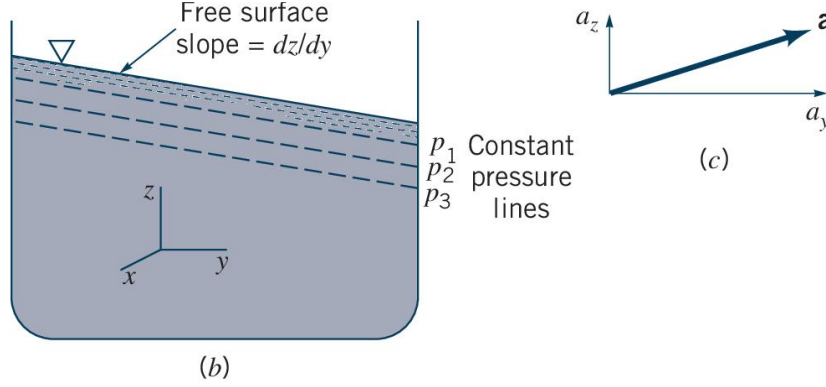
$$\Rightarrow -\frac{\partial p}{\partial x} = \rho a_x, \quad -\frac{\partial p}{\partial y} = \rho a_y, \quad -\frac{\partial p}{\partial z} = \gamma + \rho a_z$$

Problems involving fluids in motion but having no shearing stresses occur when the fluid undergoes rigid-body motion. Under rigid-body motion, fluid particles maintain fixed positions relative to one another, so there is no deformation/shearing (after some transient sloshing dies down). For example, if a container of fluid accelerates along a straight path, the fluid will move as a rigid mass, and the fluid would not deform (i.e., no shearing stresses).

By using the equation of motion for a fluid with no shearing stresses, we can determine the pressure distribution in a fluid in motion.

Linear Motion

Consider an open container of fluid that is moving with a constant acceleration a along a straight path.



Since $a_x = 0$ it implies that $\partial p / \partial x = 0$, and thus the pressure is constant along the x direction. Additionally, in the y and z directions, the pressure distribution is given by:

$$\frac{\partial p}{\partial y} = -\rho a_y$$

$$\frac{\partial p}{\partial z} = -\gamma - \rho a_z = -\rho(g + a_z)$$

The change in pressure between two closely spaced points in the fluid at (x, y, z) and $(x, y + dy, z + dz)$ is given by the first-order Taylor series expansion:

$$dp = \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz$$

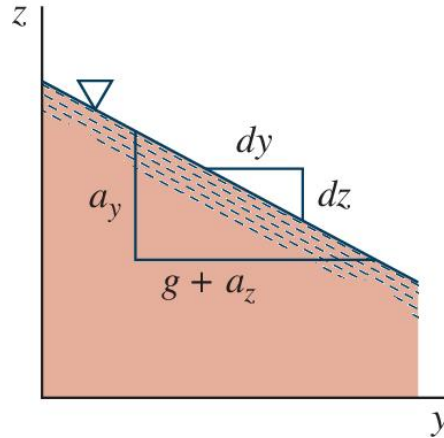
Plugging in the expressions for $\partial p / \partial y$ and $\partial p / \partial z$, we get:

$$dp = -\rho a_y dy - \rho(g + a_z) dz$$

Along a line of **constant pressure**, $dp = 0$ by definition, thus the equation above can be rearranged to give the slope of the constant pressure line:

$$\frac{dz}{dy} = -\frac{a_y}{g + a_z}$$

Note that for any fluid with a nonzero horizontal acceleration component, $a_y \neq 0$ and thus $dz/dy \neq 0$, which means that the free surface of the fluid will have a non-zero slope. Additionally, ALL lines of constant pressure will be parallel to the free surface (with the same slope).



Vertical Acceleration

For the special case of pure vertical acceleration where $a_y = 0$ and $a_z \neq 0$, the slope of the free surface is zero, and the free surface remains horizontal. However, the pressure distribution is NOT hydrostatic but is given by:

$$\frac{dp}{dz} = -\rho(g + a_z)$$

In other words, the pressure distribution is still linear but now includes a term for the vertical acceleration:

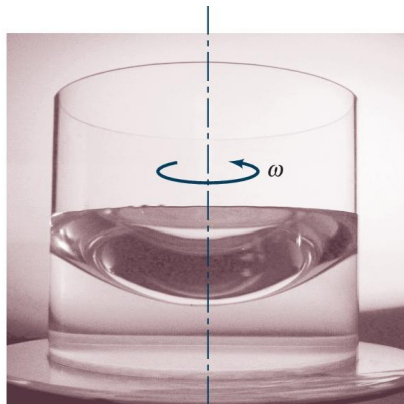
$$p = p_0 - \rho(g + a_z)z$$

Note: Pressure still increases with depth, but here z is defined as positive upwards. So, $z < 0$ below the free surface.

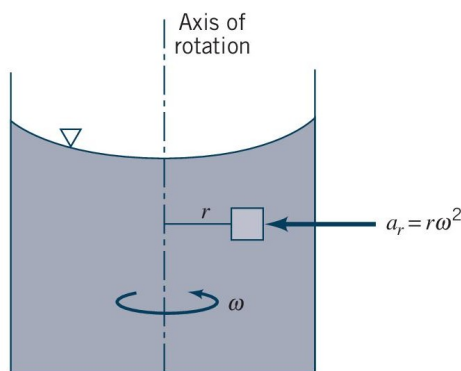
It can also be seen that if the fluid is in free fall, $a_z = -g$, then $dp/dz = 0$, and the pressure is constant throughout the fluid.

Rigid-body rotation

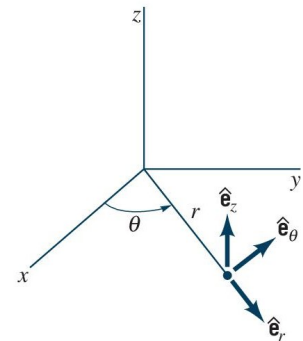
Consider a container of fluid that is rotating about a vertical axis with a constant angular velocity ω .



(a)



(b)



(c)

Since the paths of all fluid particles are circular, the acceleration of a fluid particle a distance r from the axis of rotation is given by $r\omega^2$, and it is more convenient to use cylindrical coordinates (r, θ, z) instead of Cartesian coordinates (x, y, z) . The gradient operator in cylindrical coordinates is given by:

$$\nabla = \frac{\partial}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{e}_\theta + \frac{\partial}{\partial z} \hat{e}_z$$

Thus the equation of motion for a fluid with no shearing stresses in cylindrical coordinates is given by:

$$\begin{aligned} -\nabla p - \gamma \hat{e}_z &= \rho \vec{a} \\ -\frac{\partial p}{\partial r} \hat{e}_r - \frac{1}{r} \frac{\partial p}{\partial \theta} \hat{e}_\theta - \left(\frac{\partial p}{\partial z} + \gamma \right) \hat{e}_z &= \rho a_r \hat{e}_r + \rho a_\theta \hat{e}_\theta + \rho a_z \hat{e}_z \end{aligned}$$

Since the accelerations in the cylindrical coordinate system are given by:

$$\vec{a}_r = -r\omega^2 \hat{e}_r, \quad \vec{a}_\theta = 0, \quad \vec{a}_z = 0$$

Thus, setting components equal:

$$\frac{\partial p}{\partial r} = \rho r \omega^2, \quad \frac{1}{r} \frac{\partial p}{\partial \theta} = 0, \quad \frac{\partial p}{\partial z} = -\gamma$$

Since there are two non-zero components, the pressure distribution is in terms of two variables, r and z . Again with the first order Taylor series expansion, the change in pressure between two closely spaced points in the fluid at (r, θ, z) and $(r + dr, \theta, d\theta, z + dz)$ is given by:

$$dp = \frac{\partial p}{\partial r} dr + \frac{\partial p}{\partial z} dz \implies dp = \rho r \omega^2 dr - \gamma dz$$

For a **surface of constant pressure**, $dp = 0$, and using $\gamma = \rho g$, the slope of the constant pressure surface is given by:

$$\frac{dz}{dr} = \frac{r\omega^2}{g}$$

Integrating this differential equation gives the equation of the free surface of the fluid as a parabolic surface:

$$z(r) = \frac{\omega^2}{2g} r^2 + \text{constant}$$

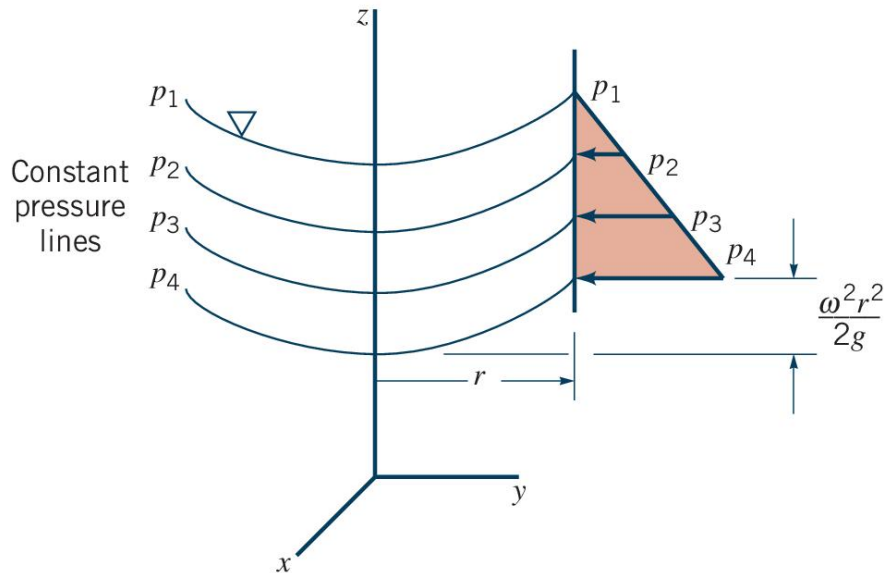
Pressure Distribution

Integrating the equation for dp gives the pressure distribution in the fluid:

$$\begin{aligned} \int dp &= \int \rho r \omega^2 dr - \int \gamma dz \\ \implies p(r, z) &= \frac{1}{2} \rho \omega^2 r^2 - \gamma z + \text{constant} \end{aligned}$$

These relations show that

- The free surface of a fluid in rigid-body rotation is parabolic.
- While the pressure distribution of a fluid in rigid-body rotation depends on both r and z , at a fixed radius r , the pressure distribution is linear with depth z .



Note: Once again note that positive z is defined as upward, so z is negative below the free surface.

Also note that when calculating the pressure *difference*, the constant of integration cancels out. Otherwise, the constant of integration can be determined by applying a boundary condition, such as the pressure at the free surface.

Elementary Fluid Dynamics—The Bernoulli Equation

3.1 Newton's Second Law

The Bernoulli equation is one of the most used and abused equations in fluid mechanics. It is derived from Newton's second law on a fluid particle as it behaves in an "ideal" manner. Note that we must respect the assumptions of the Bernoulli equation, otherwise it is not valid.

Let's define a **fluid particle** to be a small volume of fluid that is large enough to contain a sufficient number of molecules. As the fluid particle moves, it usually experiences an acceleration and thus the net force on the particle is non-zero: $\vec{F} = m\vec{a}$.

We will consider fluid motion while neglecting frictional effects due to shear stresses in a flowing fluid. To do this, we will consider a fluid that has no viscosity, called an **inviscid**

fluid. The resulting flow is called **inviscid flow**.

In reality there are no inviscid fluids, since every fluid experiences shear stresses due to viscosity. However, in many situations, the viscous effects are relatively small compared to other effects (such as inertia and pressure). Also note that the flow of gas is usually inviscid, since the viscosity of gases is very small. But always check whether the viscous effects are negligible before applying the Bernoulli equation.

For an inviscid fluid, the only forces acting on a fluid particle are body forces (gravity) and surface forces (pressure). So the net force on a fluid particle is given by:

$$(\text{Net pressure force on particle}) + (\text{Net gravity on particle}) = (\text{Particle mass}) \cdot (\text{Particle acceleration})$$

Coordinate System